

Voice-Controlled Home Bot

Offline keyword recogniser (MFCC + DTW), intent parser, device state machines, room-walking robot

AT A GLANCE

98 / 98 / 96 %

Word accuracy 20 / 10 / 5 dB

82 % raw / 99 %

grammar

Novel sentences (10 dB)

30 words / 120

Vocabulary / templates

17 (6 spoken)

Scenario commands

100 %

Spoken words heard right

149 cells / 14 trips

Robot walked

76.8 Wh in 244 s

Energy in scenario

14/14

Unit tests

VERDICT

SIM PASS

TESTS 14/14

PARSER 32 CASES

DECLARED SUBSTITUTION

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1. Objective and brief mapping

I built a smart-home simulation (four rooms; lights, five-speed fans, two thermostat-cycling ACs, a door lock, each an explicit state machine with a 230 V power-draw model), a virtual robot that walks a 16 x 10 grid map of the flat to operate each device, a pluggable speech front-end (typed text, an offline keyword recogniser I implemented on synthetic audio, and an adapter for the Google speech_recognition API), a grammar-based intent + slot parser and an actuator that executes intents with spoken/text feedback and an event log.

There is no TTS engine in the container, so the 'speech' is synthesised by me: each of the 30 vocabulary words is a fixed sequence of pseudo-phones, each phone a glottal pulse train shaped by three formants (or band-passed noise for fricatives/stops), with per-utterance pitch (95-220 Hz), tempo (0.8-1.25x), formant jitter ($\pm 7\%$) and additive white noise at a chosen SNR. Held-out test audio uses a different random seed ('voice') from the templates.

All numbers in this report were measured here by `sim/run_sim.py` and the unittest suite (`sim/metrics.json`, `scenario_log.csv`, `event_log.csv`, `transcript.txt`, `test_output.log`).

2. Architecture

`src/homebot/` is imported unchanged by `sim/run_sim.py`, `tests/`, the ROS 2 node and the Webots controller stub.

```
audio (SyntheticSpeech / mic) --> VAD segment --> MFCC(13)+delta --> DTW k-NN vs 120 templates --> b
typed text (TextInput) -----
text --> CommandParser: normalise -> split clauses (and/then/,) -> synonyms, numbers, rooms, negat
    pending confirmation for AC setpoint outside 18-28 C ; ParseError -> feedback sentence
[Intent] --> Actuator: robot.go_to(device cell) via BFS on HouseMap -> Device state machine -> Eve
Home.step(dt): AC thermal model (cooling/idle cycling), power (W) -> energy (Wh)
```

3. Implementation details

- **MFCC** (`speech.py`): pre-emphasis 0.97, 25 ms Hamming frames at 10 ms hop, 512-point $|FFT|^2$, 26 triangular mel filters (60-7000 Hz), log, DCT-II (ortho), 13 coefficients with c0 replaced by log energy, cepstral mean normalisation, plus first differences -> 26-dimensional frames.
- **DTW**: symmetric step pattern, Euclidean frame distance, Sakoe-Chiba band widened to the length difference, path-length normalised. A cheap mean/std summary prunes the 120 templates to the 60 nearest before DTW, which cut classification from 0.2 s to 0.07 s per word.
- **Templates**: 4 per word, alternating 25 dB and 12 dB SNR (multi-condition) - this single change took 5 dB accuracy from 59 % to 96 %.
- **Segmentation**: frame energy in dB smoothed over 30 ms, threshold at floor + $0.4 \times (\text{peak} - \text{floor})$ with the floor at the 10th percentile, 3-frame hangover, gaps < 50 ms merged, words < 80 ms dropped. 100 % of the 96 benchmark sentences were split into the right number of words.
- **Grammar rescoring**: the n-best (4) DTW distances per segment are combined with a bigram language model built from 39 example commands (add-0.1 smoothing) in a Viterbi pass, cost = distance - $0.6 \times \log P(w | w_{\text{prev}})$.
- **Parser**: 4 device classes with 20 synonyms (incl. 'pankha'), 4 rooms with 12 synonyms, digits and number words up to thirty-nine ('twenty two'), clause splitting on 'and / then / ,', action and room carried across clauses ('turn on the light and the fan in the hall'), negation flips the action, room inference for devices that exist in one room only (the door), context room from the previous command, fan 0-5 and AC 16-30 range checks, confirmation dialogue (yes/no) for AC setpoints outside 18-28 C, unknown-room / unknown-device / missing-value errors with a feedback sentence.
- **Devices**: Light 9 W; Fan 18/28/40/55/75 W by speed; AC 1560 W cooling, 60 W idle, thermostat hysteresis ± 0.5 C with a toy room-thermal model; door lock 0.5 W standby, 6 W for 1 s while actuating. Home integrates power to Wh every step.

4. Verification

`python3 -m unittest discover -s tests -v`: 14 tests, all passing.

Test group	What is checked	Result
Parser (4)	32 utterance cases (synonyms, number words, compounds, negation, politeness, Hinglish word, status), 6 error cases with feedback, confirmation yes/no flow, room context carry-over	pass
Devices (5)	light power, fan 0-5 transitions + limits + history, AC thermostat cools to setpoint then idles and restarts, door actuation power, home energy = 84.5 Wh in 1 h	pass
House (1)	BFS reaches every room and device cell; <code>room_of()</code> consistent with the map	pass
Recogniser (3)	MFCC shape and DTW identity, held-out 10 dB accuracy $\geq 85\%$, 5-word sentence segmented and decoded exactly	pass
End-to-end (1)	7 typed commands through parser + robot + devices, final states and event count checked	pass

5. Recogniser results (held-out voice, `sim/metrics.json`)

KEY TAKEAWAY A from-scratch MFCC + DTW recogniser reaches 98 % on held-out 10 dB words and, with a bigram grammar, 99 % on novel sentences; all 6 spoken scenario commands executed correctly. Webots / ROS 2 / Google-speech artefacts provided, not run here.

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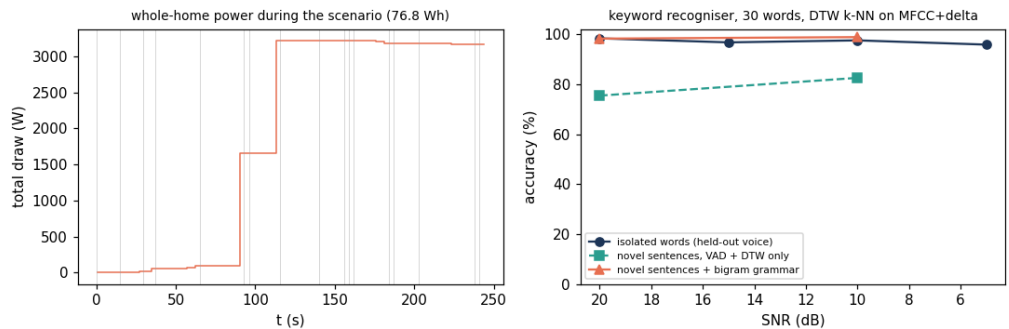
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The only isolated-word confusion at every SNR is 'to' vs 'two', which are the same phone sequence [t, u] by construction - a homophone the grammar resolves ('set fan to two' is the only place both occur). Words inside sentences are harder than isolated words (75-85 % raw) because the VAD boundaries include gap noise and clip quiet onsets; the bigram rescoring recovers almost all of it, including on the 8 novel sentences that are not in the grammar list (98 %). I report both numbers because the grammar result alone would overstate the acoustic model.

Condition	Correct	Accuracy
isolated words, 20 dB SNR (30 words x 4)	118/120	98.3 %
isolated words, 15 dB SNR (30 words x 4)	116/120	96.7 %
isolated words, 10 dB SNR (30 words x 4)	117/120	97.5 %
isolated words, 5 dB SNR (30 words x 4)	115/120	95.8 %
in-grammar sentences, 20 dB, DTW only	-	85.5 % of 117 words
in-grammar sentences, 20 dB, + bigram grammar	-	100.0 %
novel sentences, 20 dB, DTW only	-	75.4 % of 171 words
novel sentences, 20 dB, + bigram grammar	-	98.2 %
novel sentences, 10 dB, DTW only / + grammar	-	82.5 % / 98.8 %



Left: whole-home power during the scripted scenario (two ACs dominate at ~1.5 kW each). Right: recogniser accuracy vs SNR.

6. Scenario log (sim/scenario_log.csv, 17 commands: 11 typed, 6 spoken)

Summary: 13 of 17 commands executed; the other 4 were meant to be refused (two out-of-range AC setpoints that triggered a confirmation question, fan speed 9, an unknown 'microwave'). All 6 spoken commands were heard word-for-word. The robot walked 149 cells over 14 trips; 16 device events were logged; the home consumed 76.77 Wh in 243.8 s of simulated time and ended at 3160 W with both ACs cooling.

t (s)	Mode	Input (spoken words / typed text)	Heard	Bot response
0.0	text	Turn on the living room light	-	Done. Living room light on. (I walked 9 cells.)
14.8	audio	turn on the bedroom light	turn on the bedroom light	Done. Bedroom light on. (I walked 9 cells.)
29.6	audio	set bedroom fan to three	set bedroom fan to three	Done. Bedroom fan at speed 3. (I walked 3 cells.)
37.2	text	switch on the kitchen light and set the kitchen	-	Done. Kitchen light on. (I walked 15 cells.) Done. Kitchen fan at sp
65.0	audio	set bedroom ac to eighteen degrees	set bedroom ac to eighteen degrees	Done. Bedroom AC cooling at 18 C (room 31.0 C). (I walked 20 cells.)
93.0	text	set the living room ac to 16 degrees	-	16 degrees is outside the usual 18-28 range. Set the living room AC to
96.0	text	yes	-	Done. Living room AC cooling at 16 C (room 31.0 C). (I walked 13 cells)
115.6	audio	lock the door	lock the door	The hall door locked already.
140.0	text	don't turn on the hall fan	-	The hall fan off already.
156.0	text	set the kitchen fan to 9	-	Fan speed must be between 0 and 5, I heard 9.
159.0	text	turn on the microwave	-	What should I turn on in the hall?
162.0	audio	switch off kitchen light and fan	switch off kitchen light and fan	Done. Kitchen light off. (I walked 10 cells.) Done. Kitchen fan off.
183.8	text	what is on	-	living room: light on, fan off, AC cooling at 16 C (room 28.9 C) (1569
186.8	audio	unlock the door	unlock the door	Done. Hall door unlocked. (I walked 10 cells.)
202.8	text	turn off the bedroom light, then switch off the	-	Done. Bedroom light off. (I walked 16 cells.) Done. Living room ligh

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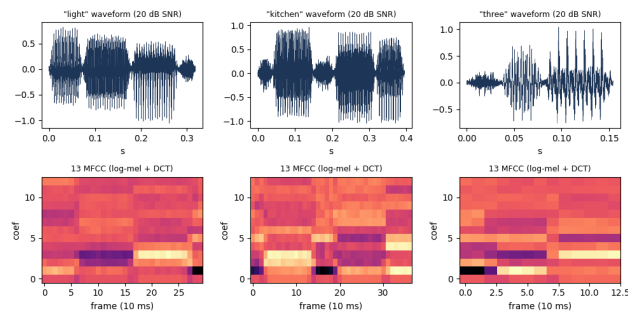
t (s)	Mode	Input (spoken words / typed text)	Heard	Bot response
237.8	text	set the bedroom ac to 30	-	30 degrees is outside the usual 18-28 range. Set the bedroom AC to 30
240.8	text	no	-	cancelled

6b. Device event log excerpt (sim/event_log.csv)

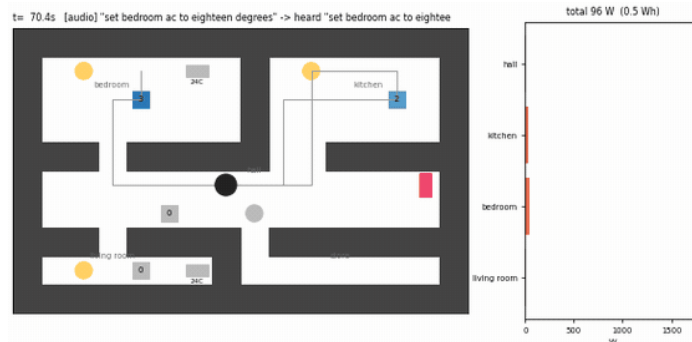
Every state change is recorded with the before/after description, so the log doubles as an audit trail; parse errors are logged too (action 'parse_error', ok = False) so that misheard or malformed commands remain visible after the fact.

t (s)	Room	Device	Action	Before	After	OK
11.8	living room	light	turn_on	light off	light on	True
26.6	bedroom	light	turn_on	light off	light on	True
34.2	bedroom	fan	set_speed	fan off	fan at speed 3	True
56.2	kitchen	light	turn_on	light off	light on	True
62.0	kitchen	fan	set_speed	fan off	fan at speed 2	True
90.0	bedroom	ac	set_temp	AC off	AC cooling at 18 C (room 31.0 C)	True
112.6	living room	ac	set_temp	AC off	AC cooling at 16 C (room 31.0 C)	True

7. Figures



Synthetic 'speech': waveforms of three vocabulary words at 20 dB SNR and their 13-coefficient MFCC sequences (the recogniser's input).



GIF frame: the robot (black) on its way to the bedroom AC after hearing 'set bedroom ac to eighteen degrees'; yellow = lights on, blue = fan speed, red bar = locked door; right: per-room power.

8. Declared substitutions / not run here

- **Google speech_recognition:** homebot.speech.GoogleSpeechAdapter implements SpeechInput over recognize_google(language='en-IN'). The package, a microphone and internet are absent here; the class reports available() == False and raises a clear error if used.
- **Webots:** worlds/home_bot.wbt (8 m x 5 m flat with the same wall layout, device marker boxes, an e-puck) and worlds/controllers/home_bot_controller/ (supervisor controller that runs the same parser/actuator and drives the e-puck along the BFS path). Webots is not installed; the controller prints a notice when imported here.
- **ROS 2 Humble:** ros2_ws/src/home_bot/ with node.py (/voice/text, /voice/audio in; /voice/transcript, /home/feedback, /home/state JSON, /robot/goal_cell, /robot/path out), package.xml, setup.py, setup.cfg, resource marker and launch file. Syntax-checked only.

9. Learnings / next steps

Two engineering lessons: multi-condition templates mattered far more than any DTW tweak, and a tiny language model on top of an honest acoustic score beat trying to make the acoustic model perfect. Next: replace the hand-made phone synthesiser with recordings of my own voice, try a small GMM/HMM per word to move beyond templates, and add speaker-adaptive cepstral normalisation so a second voice does not need new templates.